

Practicum 2

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1 Exercise 1 – Implementation of B3 and Non-Uniform Grids

1.1 Objective

The solver developed in Practicum 1 was extended to support higher-order convection discretisations and non-uniform grids. The main additions were

- the B3 second-order upwind scheme,
- the general family of λ -schemes,
- penta-diagonal matrix storage,
- the PDMA solver,
- Method A and Method B discretisations for non-uniform meshes.

The objective was to preserve compatibility with the original implementation while extending the range of available discretisation techniques.

1.2 Implementation

The original solver employed three-point stencils leading to tridiagonal systems with coefficients (a_W, a_P, a_E) . These systems were solved using the provided TDMA algorithm. To support the higher-order schemes described in Section 1.3.2, a general implementation of the λ -scheme was developed. For positive velocity, the discretisation is given by

$$\frac{d\phi}{dx} = \frac{\phi_{i+1} - \phi_{i-1}}{2h} - \lambda \frac{\phi_{i+1} - 3\phi_i + 3\phi_{i-1} - \phi_{i-2}}{h}. \quad (1)$$

Introducing a parameter λ allows a single implementation to generate the complete family of schemes:

- $\lambda = 0$: central differencing,
- $\lambda = \frac{1}{8}$: QUICK,
- $\lambda = \frac{1}{6}$: third-order λ -scheme,
- $\lambda = \frac{1}{2}$: B3.

Since these schemes involve the additional upstream node ϕ_{i-2} , the resulting matrices become penta-diagonal. Two additional diagonals were therefore introduced in the storage structure and the provided PDMA solver was used whenever a penta-diagonal system is detected. The original tridiagonal implementation remains unchanged and continues to use TDMA. For non-uniform grids, Method A and Method B were implemented following the derivations of Section 1.5. The diffusion term was discretised using the standard second-order approximation on non-uniform meshes.

1.3 Validation

The implementation of Method A and Method B was validated using the numerical benchmark of Section 1.5.2. The benchmark considers

$$L = 1, \quad U = 1, \quad k = 0.002, \quad (2)$$

on the piecewise-uniform mesh introduced in Eq. (1.21). Figures 1 and 2 show the numerical solutions obtained using the two discretisations. As can be observed, the numerical solutions reproduce exactly the

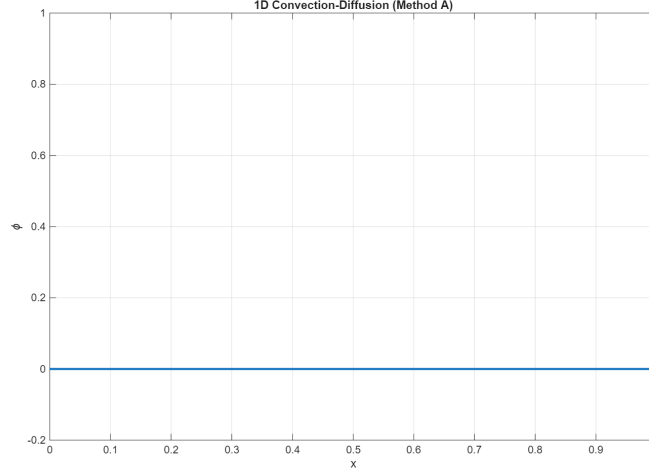


Figure 1: Numerical solution obtained with Method A.

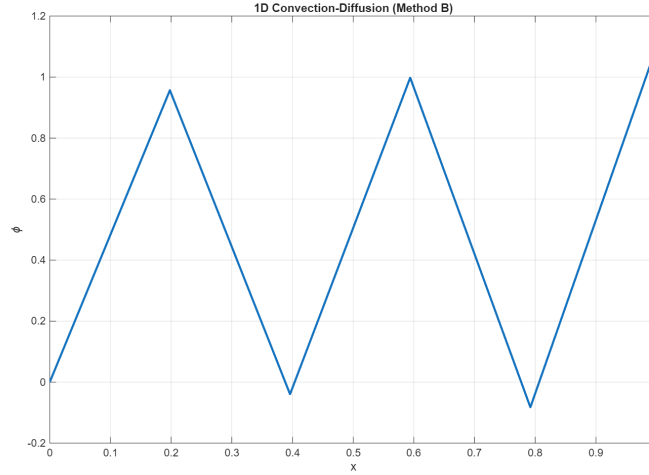


Figure 2: Numerical solution obtained with Method B.

behaviour reported in the benchmark. Method A yields the expected smooth boundary-layer profile near the outflow boundary, while Method B displays the characteristic oscillatory behaviour associated with this test case. Since the solutions coincide with the published benchmark results, the implementation of both Method A and Method B is considered correct. The implementation of the B3 scheme was validated independently on a uniform mesh. Figure 3 shows the solution obtained using the B3 discretisation. The solution exhibits the expected monotone behaviour of a second-order upwind method. Since the B3 scheme corresponds to the choice $\lambda = \frac{1}{2}$ within the general λ -scheme framework, the successful computation validates both the B3 implementation and the general λ -scheme routine.

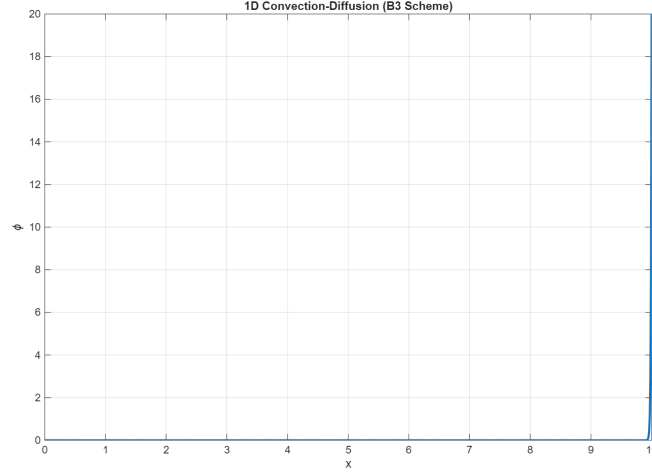


Figure 3: Solution obtained using the B3 discretisation.

1.4 Results

The benchmark problem of Section 1.5.2 was successfully reproduced. The agreement between the computed solutions and the reference results validates

- the implementation of the non-uniform mesh,
- the Method A discretisation,
- the Method B discretisation,
- the matrix assembly procedure.

In addition, the successful computation with the B3 scheme confirms that the general λ -scheme implementation is functioning correctly. Consequently, the extended solver is now able to handle both classical tridiagonal discretisations and higher-order penta-diagonal discretisations within a unified framework.

2 Exercise 2 – Theoretical Analysis

Wiggle-Free Condition for the B3 Scheme

The B3 scheme corresponds to the choice

$$\lambda = \frac{1}{2}, \quad (3)$$

in the family of λ -schemes introduced in Section 1.3.2. For positive velocity, the resulting approximation of the first derivative is

$$\frac{d\phi}{dx} = \frac{3\phi_i - 4\phi_{i-1} + \phi_{i-2}}{2h}, \quad (4)$$

which is the classical second-order upwind discretisation. Combining this approximation with the standard second-order discretisation of diffusion,

$$\frac{d^2\phi}{dx^2} = \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2}, \quad (5)$$

the steady convection–diffusion equation

$$U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad (6)$$

becomes

$$U \frac{3\phi_i - 4\phi_{i-1} + \phi_{i-2}}{2h} = k \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2}. \quad (7)$$

Introducing the mesh Péclet number

$$P = \frac{Uh}{k}, \quad (8)$$

and multiplying by h^2/k yields

$$\phi_{i+1} - \left(2 + \frac{3}{2}P\right)\phi_i + (1 + 2P)\phi_{i-1} - \frac{1}{2}P\phi_{i-2} = 0. \quad (9)$$

To investigate the occurrence of wiggles, we seek fundamental solutions of the form

$$\phi_i = r^i. \quad (10)$$

Substituting into the difference equation and dividing by r^{i-2} gives

$$r^3 - \left(2 + \frac{3}{2}P\right)r^2 + (1 + 2P)r - \frac{1}{2}P = 0. \quad (11)$$

Since a constant solution must exist, $r = 1$ is a root. Factorisation gives

$$(r - 1) \left[r^2 - \left(1 + \frac{3}{2}P\right)r + \frac{1}{2}P \right] = 0. \quad (12)$$

The remaining roots satisfy

$$r^2 - \left(1 + \frac{3}{2}P\right)r + \frac{1}{2}P = 0. \quad (13)$$

The discriminant is

$$\Delta = \left(1 + \frac{3}{2}P\right)^2 - 2P = 1 + P + \frac{9}{4}P^2. \quad (14)$$

Since

$$\Delta > 0 \quad \forall P \geq 0, \quad (15)$$

both roots are real. Furthermore, from Vieta's formulas,

$$r_1 r_2 = \frac{1}{2}P > 0, \quad r_1 + r_2 = 1 + \frac{3}{2}P, \quad (16)$$

and

$$1 + \frac{3}{2}P > \sqrt{1 + P + \frac{9}{4}P^2}, \quad (17)$$

implying

$$r_1 > 0, \quad r_2 > 0. \quad (18)$$

Hence all characteristic roots are positive:

$$r = 1, \quad r_1 > 0, \quad r_2 > 0. \quad (19)$$

Since oscillatory solutions arise when characteristic roots become negative or complex, no oscillatory mode exists for any positive value of the mesh Péclet number. Therefore the B3 scheme is wiggle-free for all mesh Péclet numbers,

$$\boxed{P \geq 0}. \quad (20)$$

This result is consistent with the general condition given in Exercise 1.3.1,

$$\lambda \geq \max\left(\frac{1}{2} - \frac{k}{Uh}, 0\right). \quad (21)$$

For the B3 scheme, $\lambda = \frac{1}{2}$, giving

$$\frac{1}{2} \geq \max\left(\frac{1}{2} - \frac{1}{P}, 0\right), \quad (22)$$

where $P = Uh/k$ is the mesh Péclet number. For $P < 2$, the right-hand side is zero, while for $P \geq 2$ the right-hand side is $\frac{1}{2} - \frac{1}{P} < \frac{1}{2}$. Therefore the inequality is satisfied for every positive value of P . Equivalently,

$$P \leq \frac{1}{\frac{1}{2} - \lambda}, \quad (23)$$

which for $\lambda = \frac{1}{2}$ gives

$$P \leq \infty. \quad (24)$$

Hence, the B3 scheme is wiggle-free for all mesh Péclet numbers,

$$0 < P < \infty. \quad (25)$$

2.1 Numerical Verification

The theoretical analysis predicts that the B3 discretisation is wiggle-free for all mesh Péclet numbers. To verify this result numerically, a series of simulations was performed using

$$U = 5, \quad k = 0.01, \quad L = 1, \quad (26)$$

with Dirichlet boundary conditions

$$\phi(0) = 0, \quad \phi(1) = 1. \quad (27)$$

A sequence of equidistant grids was considered,

$$N = \{5, 10, 20, 50, 100\}, \quad (28)$$

which results in mesh Péclet numbers

$$P = \frac{Uh}{k} = \frac{UL}{kN} = \frac{500}{N}, \quad (29)$$

namely

$$P = \{100, 50, 25, 10, 5\}. \quad (30)$$

For each grid, the steady convection–diffusion equation was solved using the B3 discretisation for convection and the standard second-order discretisation for diffusion. Figure 4 shows the numerical solutions

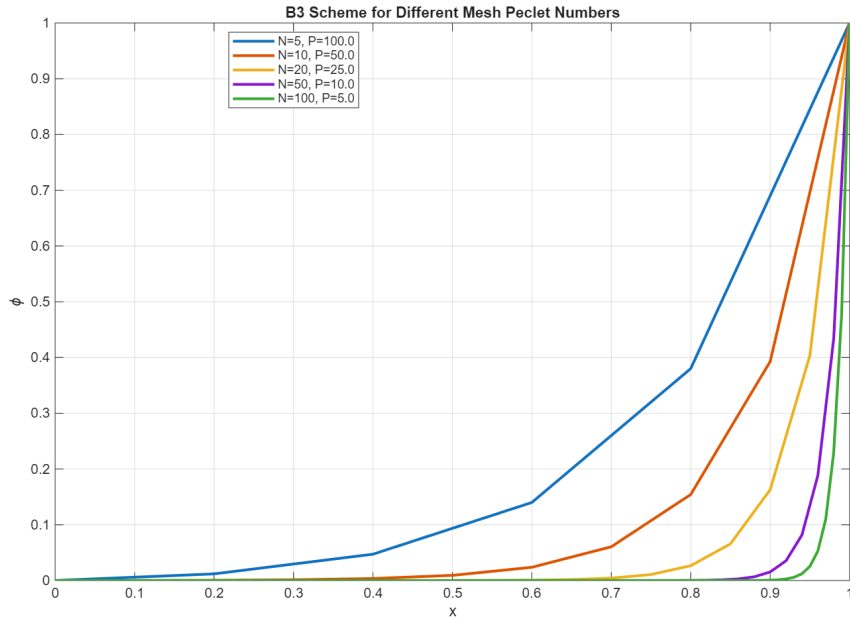


Figure 4: B3 solutions obtained for different mesh Péclet numbers.

obtained for the different mesh Péclet numbers. As the mesh is refined, the numerical solution approaches the expected boundary-layer profile near the outflow boundary. More importantly, none of the solutions exhibits oscillatory behaviour, even for the coarsest grid corresponding to

$$P = 100. \quad (31)$$

This confirms the theoretical result derived above: the B3 scheme remains monotone and wiggle-free for all mesh Péclet numbers. The numerical experiments therefore provide direct evidence that the second-order upwind discretisation successfully suppresses the oscillations that we saw in practicum 1 appears in convection-dominated problems when central differencing is used.

Should the Third-Order $\lambda = \frac{1}{6}$ Scheme be Preferred over B3?

For $\lambda = \frac{1}{6}$, the truncation error of the convective discretisation becomes

$$\tau_h = h^2 \left(\lambda - \frac{1}{6} \right) \phi_{xxx} + \mathcal{O}(h^3), \quad (32)$$

and therefore the leading second-order error term vanishes. The resulting approximation of the first derivative is thus third-order accurate. At first sight, this suggests that the $\lambda = \frac{1}{6}$ scheme should always be preferred over the B3 scheme, which is only second-order accurate. However, accuracy is not the only consideration. For practical water-wave simulations, the physical diffusion coefficient is extremely small, with $k = 10^{-6}$ and $U > 1$. Consequently, the mesh Péclet number is generally very large, even on relatively fine grids. In convection-dominated problems, robustness and monotonicity may have a bigger effect than the order of accuracy. The wiggle-free condition derived in Exercise 1.3.1 is

$$\lambda \geq \max \left(\frac{1}{2} - \frac{1}{P}, 0 \right). \quad (33)$$

For large mesh Péclet numbers ($P \gg 1$), this condition approaches $\lambda \gtrsim \frac{1}{2}$. The B3 scheme satisfies this exactly since $\lambda = \frac{1}{2}$, and is therefore wiggle-free for all mesh Péclet numbers. In contrast, for the third-order scheme $\lambda = \frac{1}{6}$, the wiggle-free condition becomes

$$\frac{1}{6} \geq \frac{1}{2} - \frac{1}{P}, \quad (34)$$

which gives $P \leq 3$. Thus, the third-order $\lambda = \frac{1}{6}$ scheme is only guaranteed to be wiggle-free when the mesh Péclet number remains below three. For practical water-wave simulations, where $k = 10^{-6}$, this condition is rarely satisfied. To illustrate this, numerical simulations were performed using $U = 5$, $k = 10^{-6}$, $L = 1$, $\phi(0) = 0$, and $\phi(1) = 1$. Three equidistant grids with $N = \{25, 50, 100\}$ were considered. The corresponding mesh Péclet numbers are $P = 2 \times 10^5$, 10^5 , and 5×10^4 , respectively. These values are several orders of magnitude larger than the theoretical wiggle-free limit $P \leq 3$ associated with the $\lambda = \frac{1}{6}$ scheme. Figure 5 compares the solutions obtained with the B3 discretisation and the third-order $\lambda = \frac{1}{6}$ scheme.

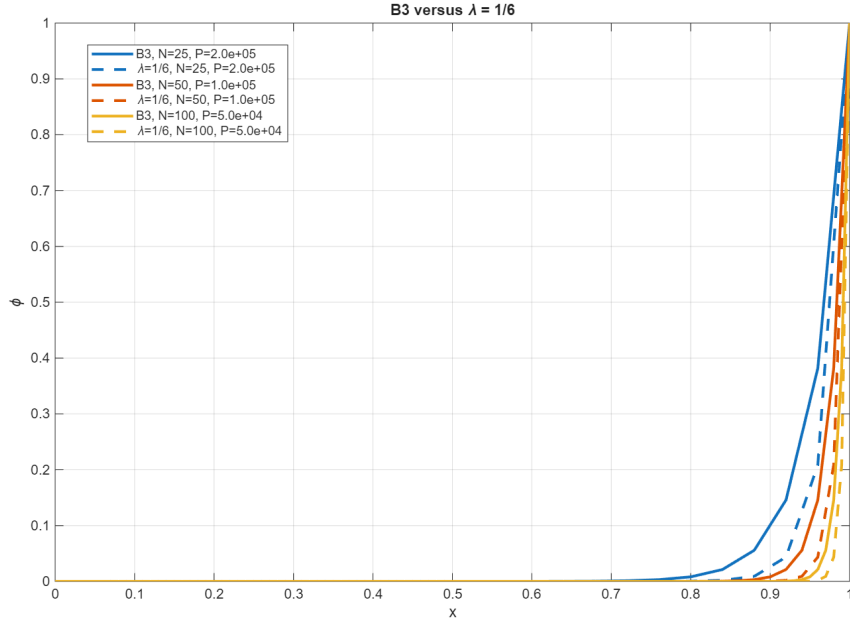


Figure 5: Comparison between the B3 scheme and the third-order $\lambda = \frac{1}{6}$ scheme for convection-dominated flow.

As expected, all solutions exhibit a strong boundary-layer character near the outflow boundary. The results also show that the additional formal order of accuracy provided by the $\lambda = \frac{1}{6}$ scheme offers

very little practical benefit for these extremely convection-dominated cases. In contrast, the B3 scheme remains monotone and satisfies the wiggle-free condition unconditionally. Consequently, although the $\lambda = \frac{1}{6}$ scheme is formally third-order accurate, the B3 discretisation can be the preferable choice for practical steady water-wave computations due to its unconditionally wiggle-free condition.

3 Exercise 3 – Accuracy Study

3.1 Verification of the B3 Order of Accuracy

To verify the formal order of accuracy of the B3 discretisation, a local truncation error study was performed using the analytical solution of Eq. (1.3). The exact solution was evaluated on a sequence of equidistant grids and its derivative was approximated using the B3 stencil

$$\left. \frac{d\phi}{dx} \right|_i \approx \frac{3\phi_i - 4\phi_{i-1} + \phi_{i-2}}{2h}. \quad (35)$$

The numerical derivative was then compared with the exact analytical derivative, and the resulting error was computed in the discrete L_2 -norm. Figure 6 shows the error as a function of the mesh size h on a log-log scale. A reference line proportional to h^2 is included for comparison. The numerical error

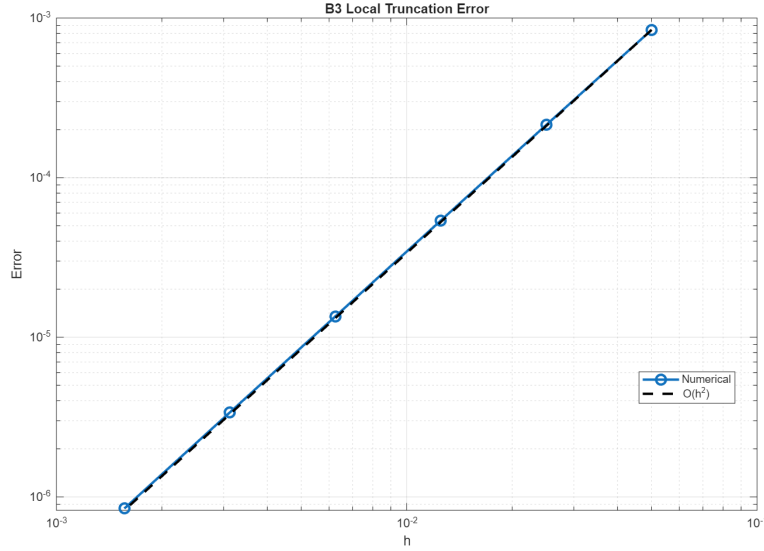


Figure 6: Local truncation error of the B3 discretisation.

follows the reference $O(h^2)$ slope almost exactly over the entire range of mesh sizes. This confirms that the truncation error decreases proportionally to h^2 , demonstrating that the B3 discretisation is indeed second-order accurate, in agreement with the theoretical result derived in Section 1.3.2.

3.2 Method A on Strongly Stretched Grids

To investigate the influence of grid stretching, Method A was applied on non-uniform meshes with

$$\frac{h_+}{h_-} = 0.95, \quad \frac{h_+}{h_-} = 0.70, \quad \frac{h_+}{h_-} = 0.50.$$

The steady convection-diffusion equation was solved using $U = 1$, $k = 0.002$, $L = 1$, together with Dirichlet boundary conditions $\phi(0) = 0$ and $\phi(1) = 1$. The numerical solutions were compared with the analytical solution given by Eq. (1.3). Figure 7 shows the numerical and analytical solutions for the three stretching ratios. For the mildly stretched mesh $\frac{h_+}{h_-} = 0.95$, the numerical solution agrees closely with the analytical solution, yielding an L_2 -error of approximately 2.2×10^{-2} . As the stretching becomes more severe, the error increases significantly. For $\frac{h_+}{h_-} = 0.70$ and $\frac{h_+}{h_-} = 0.50$, noticeable oscillations appear

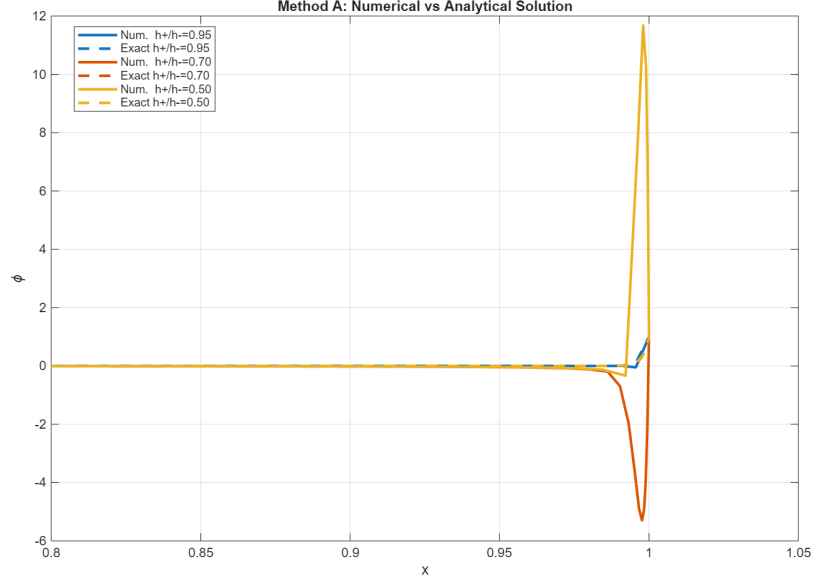


Figure 7: Comparison between the numerical and analytical solutions for Method A on stretched grids.

near the outflow boundary, indicating a loss of accuracy under strong stretching. To further investigate stability, the eigenvalues of the discretisation matrix were computed for each case. The resulting spectra are shown in Figure 8. The minimum real parts of the eigenvalues are

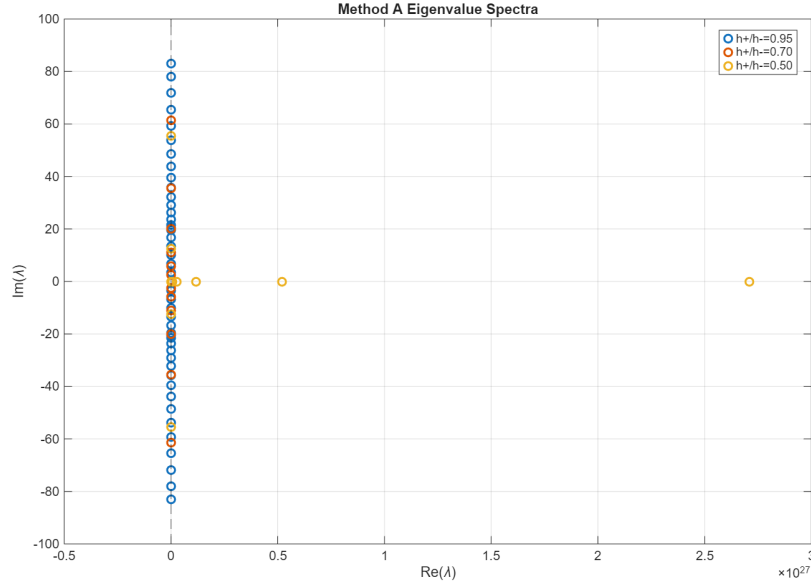


Figure 8: Eigenvalue spectra of the Method A discretisation matrix for different stretching ratios.

h_+/h_-	$\min(\text{Re}(\lambda))$
0.95	4.88
0.70	-1.72
0.50	-20.09

For $\frac{h_+}{h_-} = 0.95$, all eigenvalues remain in the stable half-plane, consistent with the good agreement observed in the solution. For stronger stretching, eigenvalues move into the unstable half-plane and

spurious oscillations appear in the numerical solution. The deterioration is particularly pronounced for $\frac{h_+}{h_-} = 0.50$. These results show that Method A performs well for mild stretching, but its accuracy and stability deteriorate as the grid becomes increasingly stretched.

3.3 Method B on Strongly Stretched Grids

The same study was repeated for Method B using the stretching ratios

$$\frac{h_+}{h_-} = 0.95, \quad \frac{h_+}{h_-} = 0.70, \quad \frac{h_+}{h_-} = 0.50.$$

The steady convection–diffusion equation was solved using $U = 1$, $k = 0.002$, and $L = 1$, together with the analytical solution of Eq. (1.3) as reference. Figure 9 compares the numerical and analytical solutions for the different stretching ratios. For the mildly stretched mesh $\frac{h_+}{h_-} = 0.95$, the numerical

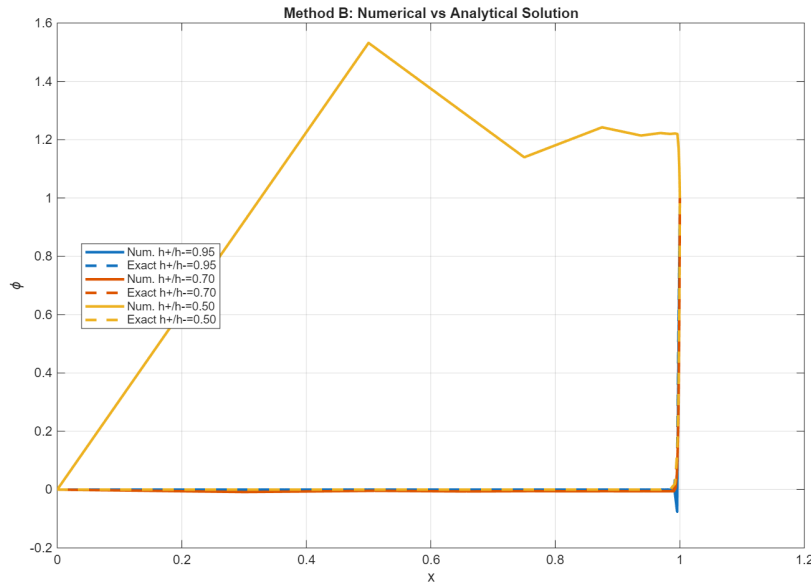


Figure 9: Comparison between the numerical and analytical solutions for Method B.

solution agrees well with the analytical solution. A similarly good agreement is observed for $\frac{h_+}{h_-} = 0.70$, which even produces the lowest error among the tested cases. For the strongly stretched mesh $\frac{h_+}{h_-} = 0.50$, however, the numerical solution becomes completely unreliable. The corresponding L_2 -error increases by more than an order of magnitude compared to the previous cases. To understand this behaviour, the eigenvalues of the discretisation matrix were computed. The resulting spectra are shown in Figure 10. The minimum real part of the spectrum is

h_+/h_-	$\min(\text{Re}(\lambda))$
0.95	4.91
0.70	1.88
0.50	-1.00

For $\frac{h_+}{h_-} = 0.95$ and $\frac{h_+}{h_-} = 0.70$, all eigenvalues remain in the stable half-plane, which is consistent with the accurate numerical solutions. For $\frac{h_+}{h_-} = 0.50$, part of the spectrum crosses into the unstable half-plane. This loss of stability coincides with the appearance of large oscillations and the breakdown of the numerical solution.

Comparison with Method A. Both methods perform well for mild stretching ($\frac{h_+}{h_-} = 0.95$). As the grid becomes more stretched, the behaviour of the two methods differs. Method A already shows

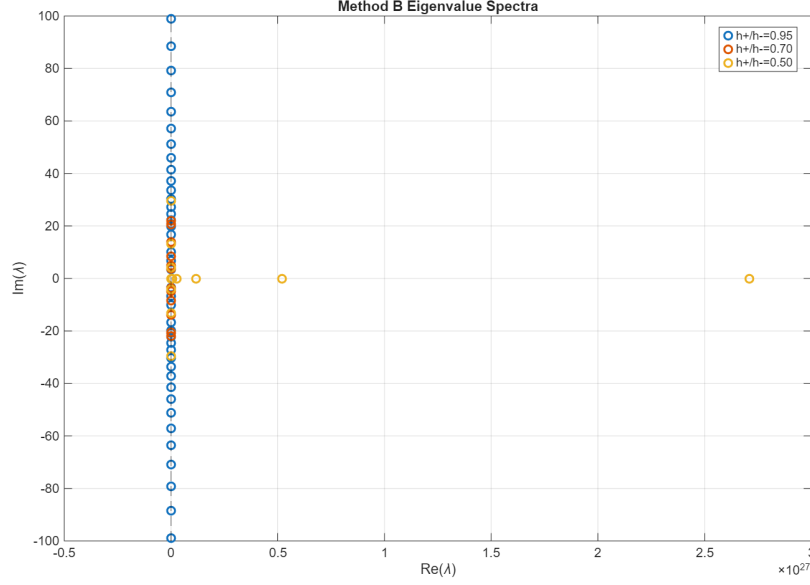


Figure 10: Eigenvalue spectra of the Method B discretisation matrix.

a significant deterioration for $\frac{h_+}{h_-} = 0.70$, with eigenvalues moving into the unstable half-plane and noticeable deviations from the analytical solution. In contrast, Method B remains accurate for this level of stretching and all eigenvalues remain in the stable half-plane. For very strong stretching ($\frac{h_+}{h_-} = 0.50$), both methods fail. In this case the numerical solutions exhibit severe oscillations and part of the spectrum lies in the unstable half-plane. The eigenvalue analysis therefore provides a direct explanation for the observed loss of accuracy: once eigenvalues cross into the unstable half-plane, non-physical oscillatory modes dominate the numerical solution.

4 Exercise 4 – Various Discretisation Methods

4.1 4.a

Figures 11–13 compare the standard upwind and smart upwind discretisations for $N = 20$ and $k = 0.05, 0.01, 0.001$. The difference between the two methods can be understood from their effective diffusion coefficients. As shown in Practicum 1, the standard upwind discretisation is equivalent to solving a modified equation with numerical diffusion

$$k_{\text{num}} = \frac{U \Delta x}{2}. \quad (36)$$

Since $U = 1$ and $\Delta x = 1/20 = 0.05$,

$$k_{\text{num}} = 0.025, \quad (37)$$

and therefore

$$k_{\text{eff}} = k + 0.025. \quad (38)$$

The smart upwind method is constructed differently. Instead of adding a fixed amount of numerical diffusion, it uses the effective diffusion coefficient

$$k_{\text{eff}} = \frac{h}{2 \tanh\left(\frac{h}{2k}\right)}, \quad (39)$$

which depends on both the mesh size and the physical diffusion coefficient. This expression can be viewed as an adaptive correction of the artificial diffusion introduced by the standard upwind method. For $k = 0.05$ ($P = 1$), we have

$$\frac{h}{2k} = \frac{0.05}{0.1} = 0.5, \quad (40)$$

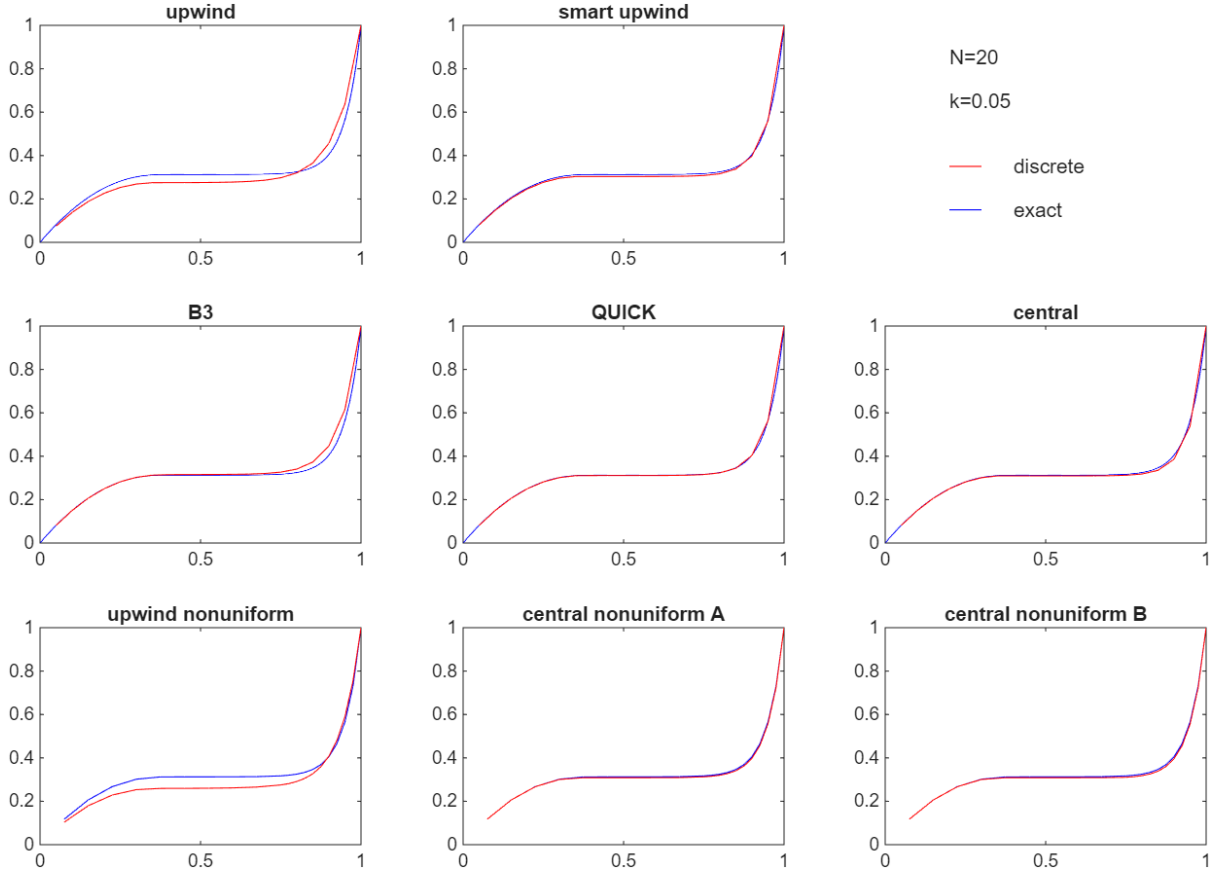


Figure 11: Solutions for $N = 20$ and $k = 0.05$ ($P = 1$).

giving

$$k_{\text{eff}} = \frac{0.025}{\tanh(0.5)} \approx 0.054. \quad (41)$$

This value is very close to the physical diffusion coefficient $k = 0.05$. Consequently, almost no artificial diffusion is added and the numerical solution is practically indistinguishable from the analytical solution. In contrast, the standard upwind method uses

$$k_{\text{eff}} = 0.05 + 0.025 = 0.075,$$

which explains the visibly more diffusive solution. For $k = 0.01$ ($P = 5$), we obtain

$$\frac{h}{2k} = 2.5, \quad (42)$$

and therefore

$$k_{\text{eff}} = \frac{0.025}{\tanh(2.5)} \approx 0.0253. \quad (43)$$

The effective diffusion is now substantially larger than the physical diffusion coefficient. As a result the smart upwind solution begins to deviate from the analytical solution. Nevertheless, it remains considerably less diffusive than the standard upwind scheme, which still uses

$$k_{\text{eff}} = 0.01 + 0.025 = 0.035.$$

Consequently, the smart upwind solution captures the boundary layer more sharply. For $k = 0.001$ ($P = 50$), we have

$$\frac{h}{2k} = 25, \quad (44)$$

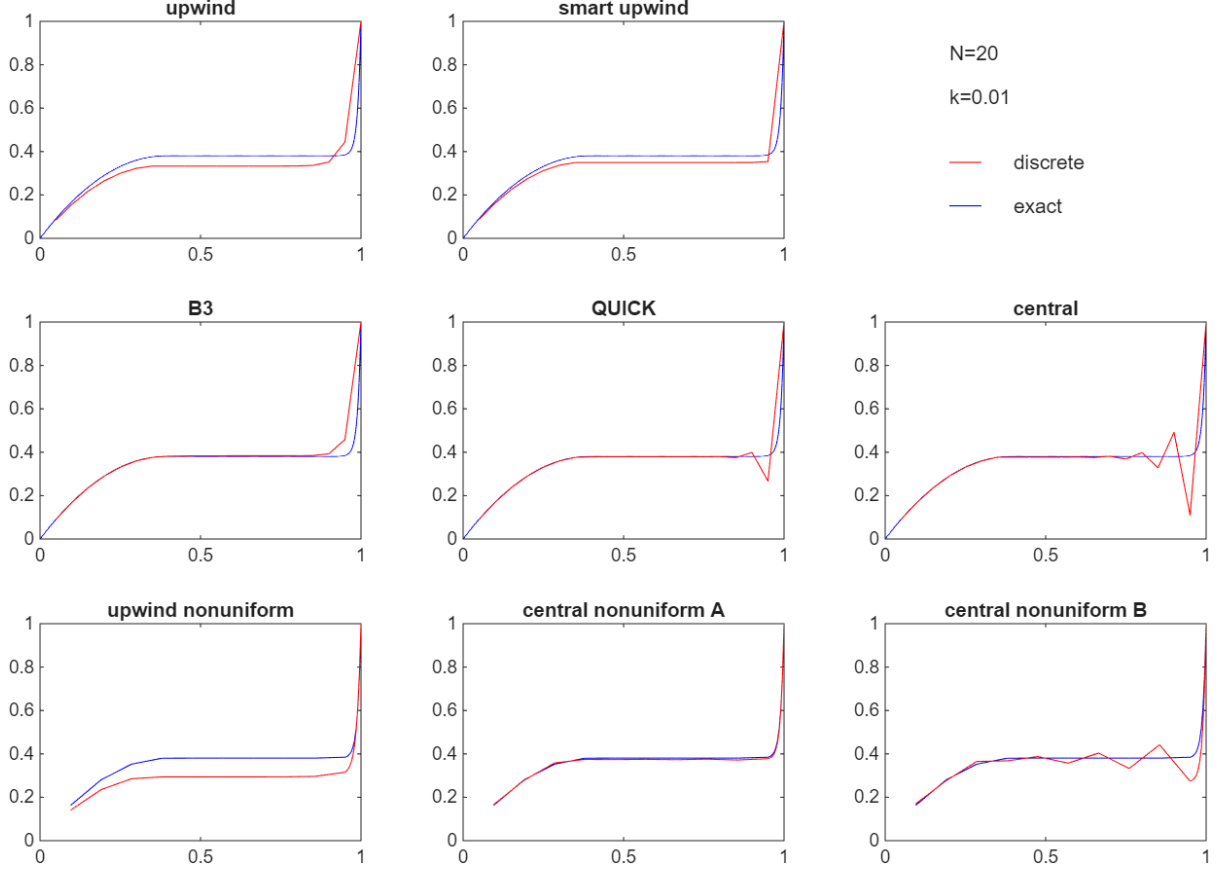


Figure 12: Solutions for $N = 20$ and $k = 0.01$ ($P = 5$).

and since

$$\tanh(25) \approx 1,$$

the smart upwind coefficient becomes

$$k_{\text{eff}} \approx \frac{h}{2} = 0.025. \quad (45)$$

Therefore, in the strongly convection-dominated limit, the smart upwind method asymptotically approaches the same effective diffusion as the standard upwind discretisation. This explains why the two numerical solutions become almost identical for $k = 0.001$. At this point the error is dominated by the coarse mesh resolution rather than by the difference between the two discretisation schemes. In summary, the smart upwind method effectively removes most of the artificial diffusion when diffusion is still significant ($k = 0.05$), partially reduces it for intermediate Péclet numbers ($k = 0.01$), and gradually approaches the classical upwind scheme in the convection-dominated limit ($k = 0.001$). This behaviour is clearly visible in Figures 11–13.

In my opinion, the smart upwind method is preferable to the standard upwind scheme in particular for diffusion-dominated regimes, due to the diffusion coefficient used. Moving towards convection-dominated regime, the boundary layer will be thinner than the grid spacing, so the error is dependency move from the artificial diffusion coefficient to the insufficient resolution, losing the smart method strength. It is possible to claim that the smart upwind method provides a clear improvement for moderate Péclet numbers while retaining the robustness of upwinding. However, for very large Péclet numbers its advantage becomes limited unless the grid is also refined.

4.2 4b

Compared to the upwind method, the λ -schemes significantly reduce artificial diffusion. The standard upwind scheme remains stable for all values of k , but produces a smoother boundary layer due to the

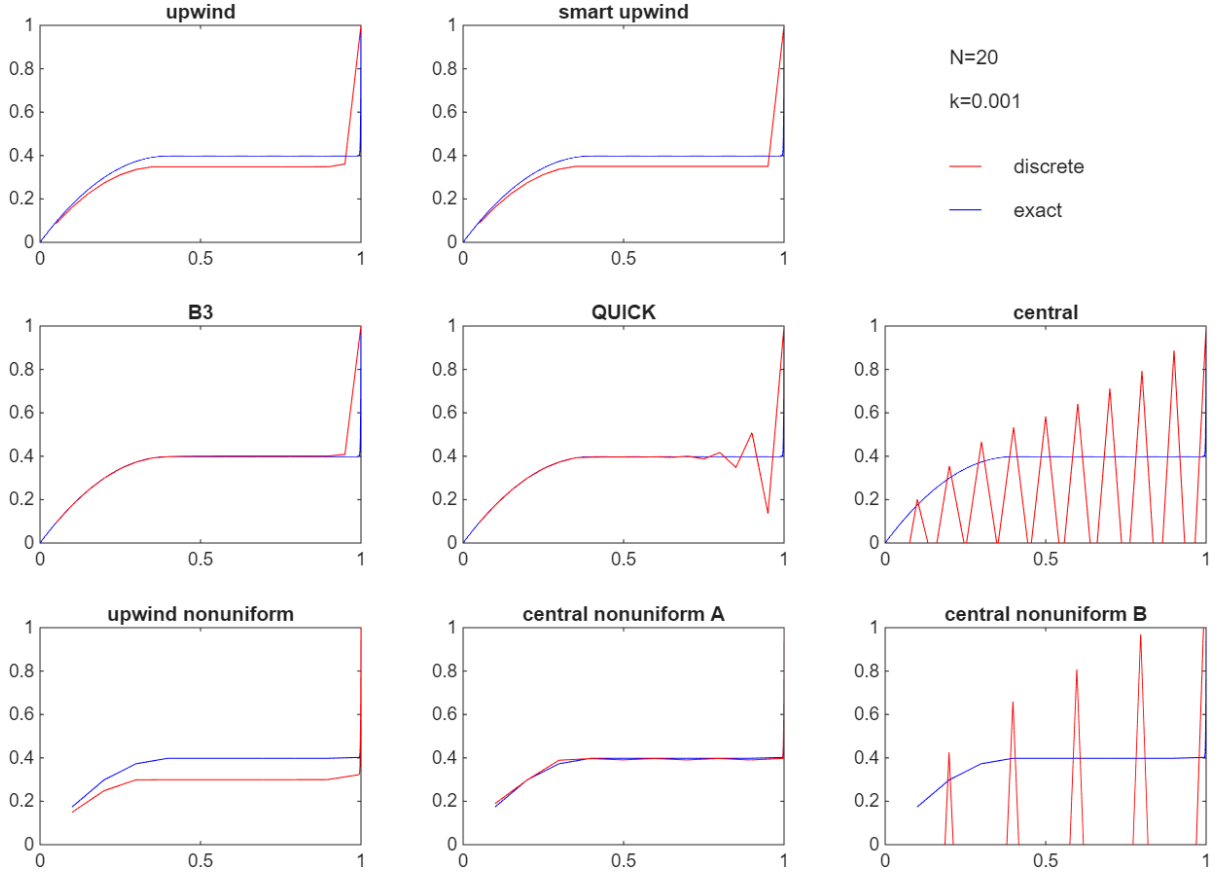


Figure 13: Solutions for $N = 20$ and $k = 0.001$ ($P = 50$).

additional numerical diffusion discussed in Practicum 1. As a result, the sharp rise near $x = 1$ is spread over several cells. The central scheme ($\lambda = 0$) eliminates this artificial diffusion and therefore gives the sharpest approximation of the exact solution. However, this comes at the price of reduced stability: oscillations become visible as the mesh Péclet number increases. This behaviour is already visible for $k = 0.015$ and becomes more pronounced as k decreases. The QUICK scheme ($\lambda = \frac{1}{8}$) provides a compromise between accuracy and robustness. For $k = 0.05$, the solution is smooth and captures the boundary layer much better than the upwind method. As k decreases, oscillations gradually appear near the outlet. The additional computations performed for $k = 0.019$ and $k = 0.015$ show that the QUICK solution remains essentially wiggle-free for $k = 0.019$, while small oscillations are already visible for $k = 0.015$. Therefore, the wiggle boundary lies approximately between these two values. The B3 scheme ($\lambda = \frac{1}{2}$) combines the advantages of both approaches. It remains wiggle-free while introducing considerably less artificial diffusion than the standard upwind method. As a result, it provides the best overall compromise between stability and accuracy among the tested schemes. Empirically, the QUICK method starts to wiggle at approximately

$$k_{\text{crit}} \approx 0.018.$$

This agrees with the theoretical wiggle criterion for the λ -schemes,

$$\lambda \geq \max\left(\frac{1}{2} - \frac{1}{P}, 0\right), \quad (46)$$

where $P = Uh/k$ is the mesh Péclet number. For QUICK,

$$\lambda = \frac{1}{8},$$

which yields

$$P \leq \frac{8}{3}.$$

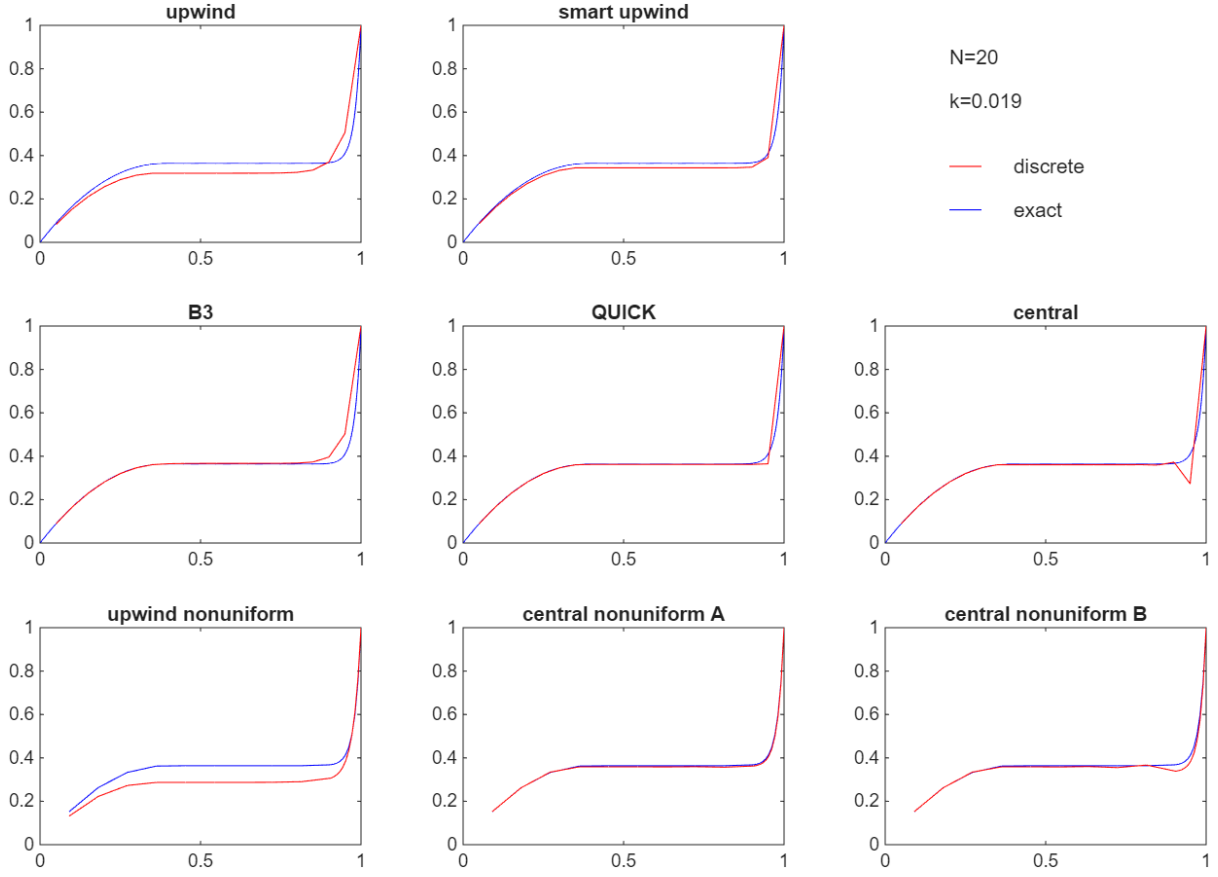


Figure 14: Solutions for $N = 20$ and $k = 0.019$. The QUICK solution remains essentially wiggle-free.

Using

$$U = 1, \quad h = \frac{1}{20} = 0.05,$$

gives

$$\frac{0.05}{k_{\text{crit}}} = \frac{8}{3},$$

and therefore

$$k_{\text{crit}} = 0.01875.$$

The theoretical prediction $k_{\text{crit}} = 0.01875$ lies precisely between the numerical tests performed at $k = 0.019$ and $k = 0.015$. The observed onset of oscillations is therefore in excellent agreement with the theoretical wiggle boundary.

4.3 4c

The results on the stretched grid show clear differences between the upwind method and the two non-uniform central discretisations, Method A and Method B. The upwind solution remains smooth and free of oscillations, but its accuracy is not improved by the grid stretching. The reason is that the dominant error of the upwind scheme is the artificial diffusion introduced by the discretisation itself. Although the mesh is refined near the outlet boundary, a significant part of the domain is still covered by relatively coarse cells. The upwind scheme therefore continues to solve a problem with an effective diffusion coefficient larger than the physical diffusion coefficient, resulting in an overly smooth solution and a smeared boundary layer. Methods A and B, on the other hand, are based on central differencing and therefore do not introduce the same amount of artificial diffusion. Consequently, both methods capture the boundary layer much more sharply than the upwind scheme. The plateau value is also reproduced more accurately. However, the two central formulations behave differently on strongly stretched meshes. Method A remains relatively robust and produces a smooth solution that agrees well with the exact

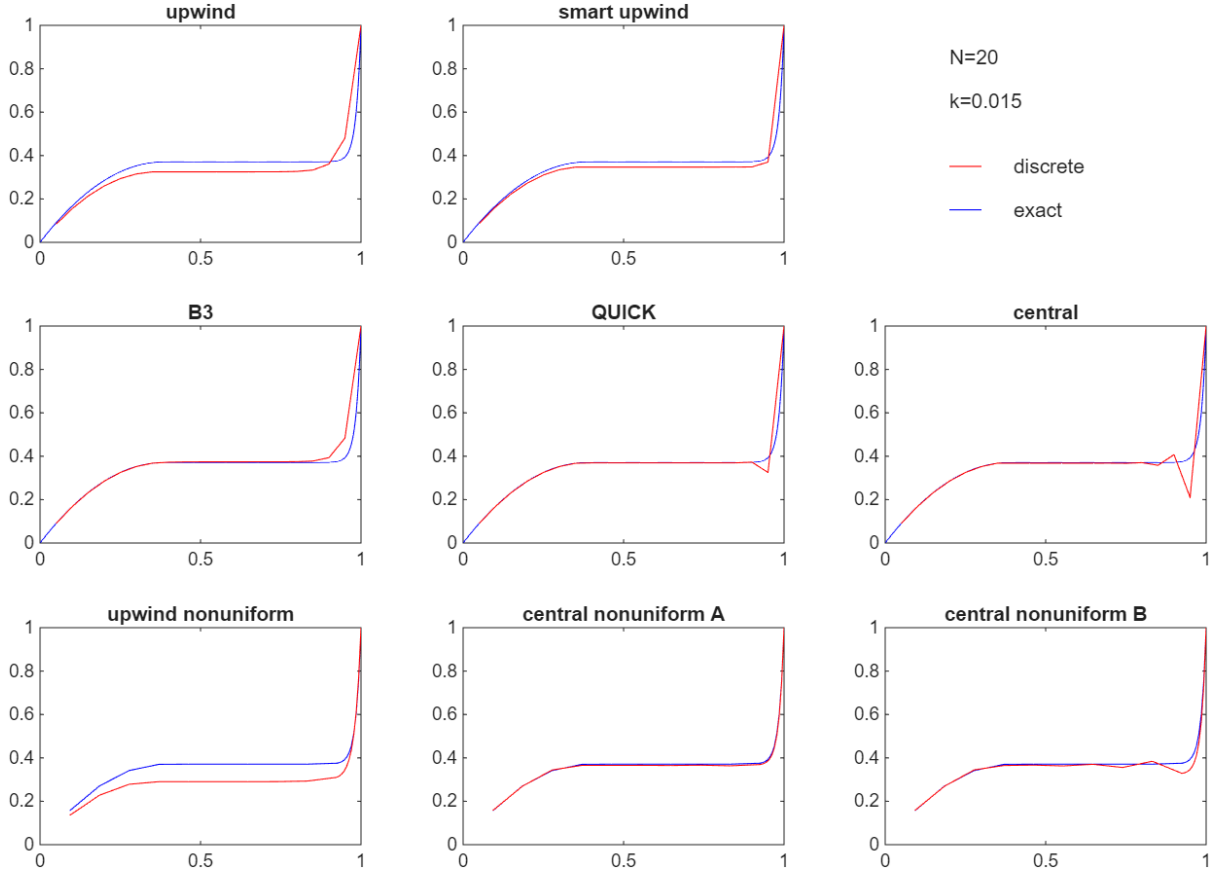


Figure 15: Solutions for $N = 20$ and $k = 0.015$. Small oscillations become visible in the QUICK solution.

solution. Method B is more sensitive to the grid stretching and may develop oscillations when the stretching becomes sufficiently severe. This behaviour is consistent with the eigenvalue analysis discussed previously, where Method B was shown to lose stability more readily than Method A.

4.4 4d

To investigate the stability of Method B, the diffusion coefficient k was gradually reduced while monitoring the eigenvalues of the coefficient matrix. The results show a clear relation between the location of the eigenvalues and the appearance of oscillations in the numerical solution. For $k = 0.05$, all eigenvalues are real and positive, with the smallest eigenvalue equal to

$$\lambda_{\min} = 6.37.$$

The numerical solution is smooth and agrees well with the analytical solution. For $k = 0.019$, all eigenvalues still have positive real parts. The smallest real part is

$$\min(\text{Re}(\lambda)) = 4.74,$$

and no wiggles are visible in the numerical solution. This case is shown in Figure 14. Reducing the diffusion coefficient further to $k = 0.015$ produces small oscillations in the numerical solution. The corresponding eigenvalues now cluster much closer to the imaginary axis, with the smallest real part approximately

$$\min(\text{Re}(\lambda)) = 3.61.$$

Although all eigenvalues are still located in the stable half-plane, the system is already becoming sensitive to oscillatory modes. This is therefore the first value of k for which wiggles are clearly observed in the numerical solution. When k is reduced to $k = 0.01$, several eigenvalues lie extremely close to the imaginary axis,

$$\lambda = 2.30 \pm 9.93i,$$

after accounting for the 10^{-3} scaling shown in the MATLAB output. At this point the oscillations become clearly visible. Finally, for $k = 0.001$ one eigenvalue becomes negative,

$$\lambda = -5.61 \times 10^{-3}.$$

This indicates that an eigenvalue has crossed the imaginary axis and entered the unstable half-plane. The numerical solution then exhibits very large, non-physical oscillations. Therefore, based on the numerical experiments, the crossing of the imaginary axis occurs between

$$0.001 < k_{\text{cross}} < 0.01,$$

while the onset of visible wiggles is already observed at approximately

$$k \approx 0.015.$$

The matrix becomes closest to singular for $k = 0.001$, where the smallest eigenvalue is very close to zero. In this regime the solution is dominated by the eigenmode associated with this small eigenvalue, causing large oscillations throughout the domain. The numerical solution no longer resembles the analytical solution and is therefore not physically meaningful. These results demonstrate a clear connection between the eigenvalue spectrum and the quality of the numerical solution. As the smallest eigenvalue approaches zero, the matrix becomes increasingly ill-conditioned, oscillations appear, and eventually the discrete solution becomes dominated by spurious modes.